



Rigidity of complete Kähler–Einstein metrics under cscK perturbations



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ABSTRACT

In this paper, we study constant scalar curvature Kähler (cscK) metrics on complete non-compact Kähler–Einstein manifolds. We give sufficient conditions under which a cscK perturbation of a Kähler–Einstein metric must remain Kähler–Einstein. As a model case, we prove that the Bergman metric on a bounded strictly pseudoconvex domain is Kähler–Einstein whenever it has constant scalar curvature. In particular, combined with Huang–Xiao’s resolution of Cheng’s conjecture, this yields the ball characterization for smooth bounded strictly pseudoconvex domains.

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1. Introduction

1.1. Canonical metrics

A classical problem in Kähler geometry is to find a canonical representative in a given Kähler class $[\omega]$. Kähler–Einstein metrics are among the most natural and intensively studied ones. A Kähler metric ω is called Kähler–Einstein if the Ricci curvature is proportional to the metric, that is

$$\text{Ric}(\omega) = \lambda\omega,$$

for some $\lambda \in \mathbb{R}$. After rescaling the metric, one may normalize the Einstein constant to be $\lambda = 1$, 0 or -1 .

On compact Kähler manifolds, the existence of Kähler–Einstein metrics depends on the sign of the first Chern class $c_1(M)$:

- (1) If $c_1(M) < 0$, M has an ample canonical bundle, a problem solved by Aubin [1] and Yau [37];
- (2) If $c_1(M) = 0$, M is Calabi–Yau, as solved by Yau [37];

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- (3) If $c_1(M) > 0$, M is Fano, and there are classical obstructions to the existence of Kähler–Einstein metrics, going back to Matsushima [24], Futaki [18], and Tian [32]. In a series of papers, Chen–Donaldson–Sun [9–11] proved that K -polystability of a Fano manifold is sufficient for the existence of a Kähler–Einstein metric, confirming the “ K -polystability $\Rightarrow$ KE” direction of the Yau–Tian–Donaldson conjecture.

For the non-compact case, significant progress on the existence of the Kähler–Einstein metric has been made by Cheng–Yau [14], Mok–Yau [25], Tian–Yau [33,34], and Guedj–Kolev–Yeganehfar [20].

More generally, on a compact Kähler manifold, a Kähler–Einstein metric in a Kähler class $[\omega]$ necessarily satisfies $2\pi c_1(M) = \lambda[\omega]$ for some $\lambda \in \mathbb{R}$. When this cohomological condition fails, constant scalar curvature Kähler (often abbreviated as cscK) metrics provide the natural broader class of canonical metrics. On compact Kähler manifolds, the average of the scalar curvature $\hat{R}$ is given by

$$\hat{R} = \frac{2n\pi c_1(M) \cup [\omega]^{n-1}}{[\omega]^n}$$

which depends only on the Kähler class $[\omega]$.

1.2. YTD conjecture for the cscK metric

For a polarized manifold (M, L) , the Yau–Tian–Donaldson conjecture predicts that the existence of a unique cscK metric in $c_1(L)$ should be equivalent to the uniform K -stability of (M, L) , linking the K -energy’s analytic behavior to algebraic stability via test configurations [16,32,38].

In [12], Chen proposed a new continuity path for the cscK equation and proved the openness along this path. Subsequently, Chen and Cheng [7,8] established the decisive a priori estimates and proved existence under the properness of the K -energy. We refer the reader to [2–6,17,30] for further developments.

1.3. CscK metrics on complete noncompact manifolds

Despite the substantial progress in the compact case, cscK metrics on complete non-compact Kähler manifolds remain much less understood. A basic analytic difficulty is the failure of a global $\partial\bar{\partial}$ -lemma in the non-compact setting, so one cannot in general work within a fixed Kähler class in the usual compact sense. For this reason, throughout the paper we fix a background complete Kähler metric ω and restrict attention to perturbations of the form $\omega_\varphi = \omega + \sqrt{-1}\partial\bar{\partial}\varphi$ with potential φ . Together with the (a priori unknown) log volume ratio F , the coupled cscK equation reads

$$\begin{cases} \omega_\varphi^n = e^F \omega^n, \\ \Delta_{\omega_\varphi} F = -\hat{R} + \text{tr}_{\omega_\varphi} \text{Ric}(\omega), \end{cases} \quad (1.1)$$

whose solvability on suitable complete manifolds yields a cscK metric of scalar curvature $\hat{R}$. Note that Kähler–Einstein metrics also solve (1.1). On compact manifolds, if a cscK metric is cohomologous to a Kähler–Einstein metric, then it must itself be Kähler–Einstein. By contrast, in the complete non-compact case there is no general rigidity principle forcing cscK perturbations to coincide with the background Kähler–Einstein metric.

On a negatively curved Kähler–Einstein manifold (M, ω) , we normalize by $\text{Ric}(\omega) = -\omega$. For cscK perturbations with scalar curvature $\hat{R} = -n$, the coupled system (1.1) on (M, ω) reduces to

$$\omega_\varphi^n = e^F \omega^n, \quad \Delta_{\omega_\varphi} F = n - \text{tr}_{\omega_\varphi} \omega,$$

which is the form used throughout the paper.

Question 1.1. Can one give a *sufficient condition* under which every cscK perturbation of a complete Kähler–Einstein metric is again Kähler–Einstein?

Question 1.1 admits an affirmative answer under either of two sets of verifiable hypotheses: a parabolic condition, and a quantitative non-parabolic condition combining a positive spectral gap, a C^2 -bound and an annular L^2 -growth bound on the potential φ . Specifically, our main result is the following.

Theorem 1.2. *Let (M^n, ω) be a complete Kähler–Einstein manifold with negative scalar curvature $R(\omega) = -n$. Suppose there exists $\varphi \in C^\infty(M)$ normalized with $\sup_M \varphi = 0$ such that $\omega_\varphi = \omega + \sqrt{-1}\partial\bar{\partial}\varphi$ defines a complete cscK metric satisfying $R(\omega_\varphi) = -n$. Assume that $\text{Ric}(\omega_\varphi)$ is bounded from below and that there exists a constant $\lambda > 0$ such that $\omega_\varphi^n \geq \lambda\omega^n$. Then ω_φ is Kähler–Einstein whenever one of the following conditions holds:*

- (i) (M, ω) is parabolic;
- (ii) (M, ω) is non-parabolic, $\|\varphi\|_{C^2(M, \omega)} < \infty$ and there exist a point $p \in M$, constants $C_0 > 0$ and $0 < \delta < 2\sqrt{\lambda_1(-\Delta_{\omega_\varphi})}$ such that, for all $R \geq 1$,

$$\int_{B_{\omega_\varphi}(p, 2R) \setminus B_{\omega_\varphi}(p, R)} |\varphi|^2 \omega_\varphi^n \leq C_0 e^{2\delta R}.$$

Remark 1.3 (*Calabi–Yau case*). When the background metric is Calabi–Yau, decay of the logarithmic volume ratio at infinity already forces a scalar-flat perturbation to be Ricci-flat (see Proposition 4.2).

Recall that a complete Riemannian manifold is *non-parabolic* if it admits a positive minimal Green’s function, and *parabolic* otherwise. In the present setting, Lemma 3.1 implies that the normalized potential φ is ω -subharmonic. This is sufficient to prove rigidity in the parabolic case. In the quantitative non-parabolic case, the natural analytic quantity is instead $u := F - \varphi$, which is ω_φ -harmonic. Nevertheless, Lemma 4.8 shows that, under the C^2 -bound on φ , the annular L^2 -growth conditions for u and φ are equivalent. We therefore formulate Theorem 1.2 (ii) directly in terms of the potential φ . Moreover, Lemma 3.3 shows that ω_φ and ω are uniformly equivalent. Consequently, $\lambda_1(-\Delta_{\omega_\varphi}) > 0$ if and only if $\lambda_1(-\Delta_\omega) > 0$. The precise growth threshold in Theorem 1.2 (ii) is written using ω_φ , since both the numerical value of λ_1 and the radial variable in the annular condition depend on the chosen metric.

1.4. Bounded strictly pseudoconvex domains as a model case

As a concrete model case of Question 1.1, we next consider bounded strictly pseudoconvex domains, where the rigidity problem admits a particularly geometric form. Let $\Omega \subset \mathbb{C}^n$ be a bounded strictly pseudoconvex domain with smooth boundary. In [14], Cheng and Yau proved that there is a unique complete Kähler–Einstein metric on Ω constructed by a global strictly plurisubharmonic defining function. This leads to the following natural question: *Can a bounded strictly pseudoconvex domain carry a complete cscK metric that is not Kähler–Einstein?*

On a bounded strictly pseudoconvex domain, two natural types of complete Kähler metrics arise. The first is the Bergman metric ω_B . The second consists of the metrics ω_ρ associated with strictly plurisubharmonic defining functions ρ . Any two metrics of the latter type differ by a global $\sqrt{-1}\partial\bar{\partial}$ -term. We refer to Section 5 for the relevant background and details. For these two types of metrics, we prove the following rigidity statements.

Theorem 1.4. *Let $\Omega \subset \mathbb{C}^n$ be a bounded strictly pseudoconvex domain with C^2 -boundary.*

- (1) If ω_ρ is cscK and $\rho \in C^\infty(\Omega) \cap C^4(\bar{\Omega})$ satisfies $J(\rho) = 1$ on $\partial\Omega$ where J is the Fefferman operator, then ω_ρ coincides with the unique Kähler–Einstein metric constructed by Cheng–Yau.
- (2) If the Bergman metric ω_B is cscK, then ω_B is Kähler–Einstein. Moreover, if $\partial\Omega \in C^\infty$, then Ω is biholomorphic to the unit ball.

Theorem 1.4 (2) may be viewed as a cscK extension of Huang–Xiao’s rigidity theorem for Kähler–Einstein Bergman metrics [22, Theorem 1]. Indeed, once the Bergman metric is assumed to have constant scalar curvature, our rigidity result forces it to be Kähler–Einstein. Then Huang–Xiao’s result yields the ball characterization on bounded smooth strictly pseudoconvex domains.

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2. Preliminaries

2.1. Notations

Let (M^n, ω) be an n -dimensional complete Kähler manifold. In local holomorphic coordinates $(z^1, \dots, z^n)$, we write

$$\omega = \sqrt{-1} g_{i\bar{j}} dz^i \wedge d\bar{z}^j,$$

where we adopt the Einstein summation convention. We use ∂_i and $\partial_{\bar{j}}$ as shorthand for $\frac{\partial}{\partial z^i}$ and $\frac{\partial}{\partial \bar{z}^j}$, respectively, and denote by $(g^{i\bar{j}})$ the inverse matrix of $(g_{i\bar{j}})$. Given two tensors A and B of the same type, their Hermitian inner product with respect to ω is denoted by $\langle A, B \rangle_\omega$, and the corresponding norm by $|A|_\omega$.

The Ricci form is given by

$$\text{Ric}(\omega) = \sqrt{-1} R_{i\bar{j}} dz^i \wedge d\bar{z}^j, \quad R_{i\bar{j}} = -\partial_i \partial_{\bar{j}} \log \det(g_{k\bar{l}}),$$

and the scalar curvature is

$$R(\omega) = \text{tr}_\omega \text{Ric}(\omega) = g^{i\bar{j}} R_{i\bar{j}}.$$

For $f \in C^\infty(M, \mathbb{R})$, the gradient vector field is

$$\nabla f = \frac{1}{2} g^{i\bar{j}} (\partial_i f \partial_{\bar{j}} + \partial_{\bar{j}} f \partial_i),$$

and the Laplacian is defined by the trace of the complex Hessian $\sqrt{-1} \partial \bar{\partial} f$ with respect to ω , namely

$$\Delta_\omega f := \text{tr}_\omega (\sqrt{-1} \partial \bar{\partial} f) = g^{i\bar{j}} \partial_i \partial_{\bar{j}} f.$$

2.2. Omori–Yau maximum principle

We record the generalized maximum principle on complete Kähler manifolds in the two forms used later in the paper.

Proposition 2.1 (Omori [28], Yau [36]). Let (M, ω) be a complete Kähler manifold, and let $u \in C^2(M)$ satisfy $\sup_M u < +\infty$.

(i) If the sectional curvature of ω is bounded from below, then there exists a sequence $\{z_k\} \subset M$ such that

$$\lim_{k \rightarrow \infty} u(z_k) = \sup_M u, \quad \lim_{k \rightarrow \infty} |\nabla u(z_k)|_\omega = 0, \quad \limsup_{k \rightarrow \infty} \sqrt{-1} \partial \bar{\partial} u(z_k) \leq 0,$$

where the last inequality is understood in the sense of Hermitian forms.

(ii) If the Ricci curvature of ω is bounded from below, then there exists a sequence $\{z_k\} \subset M$ such that

$$\lim_{k \rightarrow \infty} u(z_k) = \sup_M u, \quad \lim_{k \rightarrow \infty} |\nabla u(z_k)|_\omega = 0, \quad \limsup_{k \rightarrow \infty} \Delta_\omega u(z_k) \leq 0.$$

3. Basic estimates for cscK metrics

Let (M^n, ω) be a complete Kähler–Einstein manifold with negative scalar curvature, normalized by $R(\omega) = -n$. Suppose $\tilde{\omega}$ is a complete cscK metric on M with

$$R(\tilde{\omega}) = R(\omega) = -n.$$

Then there exists a smooth function F such that

$$\tilde{\omega}^n = e^F \omega^n, \tag{3.1}$$

and

$$\Delta_{\tilde{\omega}} F = n - \operatorname{tr}_{\tilde{\omega}} \omega. \tag{3.2}$$

We can verify that

$$\begin{aligned} R(\tilde{\omega}) &= \operatorname{tr}_{\tilde{\omega}} \operatorname{Ric}(\tilde{\omega}) \\ &= -\operatorname{tr}_{\tilde{\omega}} \sqrt{-1} \partial \bar{\partial} \log \left(\frac{\tilde{\omega}^n}{\omega^n} \right) + \operatorname{tr}_{\tilde{\omega}} \operatorname{Ric}(\omega) \\ &= -\Delta_{\tilde{\omega}} F - \operatorname{tr}_{\tilde{\omega}} \omega \\ &= -n. \end{aligned}$$

Throughout this section, we work under the above normalization.

We first present a basic comparison for the volume form of a cscK metric.

Lemma 3.1. Let (M, ω) be a complete Kähler–Einstein manifold, and let $\tilde{\omega}$ be a complete cscK metric with $R(\tilde{\omega}) = R(\omega) = -n$. If $\operatorname{Ric}(\tilde{\omega})$ is bounded from below and the log volume ratio F is bounded from below, then

$$\tilde{\omega}^n \geq \omega^n. \tag{3.3}$$

Moreover, if equality holds at any point $p \in M$, then $\tilde{\omega} = \omega$ on M .

Proof. Writing $u = \omega^n / \tilde{\omega}^n > 0$, we compute

$$\begin{aligned}\Delta_{\tilde{\omega}} \log u &= \Delta_{\tilde{\omega}} \log (\omega^n / \tilde{\omega}^n) \\ &= \operatorname{tr}_{\tilde{\omega}} \omega - n \\ &\geq nu^{\frac{1}{n}} - n,\end{aligned}$$

where the last inequality follows from the arithmetic-geometric mean inequality.

Then we obtain

$$\Delta_{\tilde{\omega}} \log u = \frac{\Delta_{\tilde{\omega}} u}{u} - \frac{|\nabla u|_{\tilde{\omega}}^2}{u^2} \geq nu^{\frac{1}{n}} - n,$$

which leads to

$$\Delta_{\tilde{\omega}} u - \frac{|\nabla u|_{\tilde{\omega}}^2}{u} \geq nu^{\frac{1}{n}+1} - nu.$$

Since u is bounded from above and $\operatorname{Ric}(\tilde{\omega})$ is bounded from below, the generalized maximum principle applied to u with respect to $\tilde{\omega}$ yields a sequence $\{z_\alpha\} \subset M$ such that

$$0 \geq \limsup_{\alpha \rightarrow \infty} \Delta_{\tilde{\omega}} u(z_\alpha) \geq n \lim_{\alpha \rightarrow \infty} u^{\frac{1}{n}+1}(z_\alpha) - n \lim_{\alpha \rightarrow \infty} u(z_\alpha).$$

This implies $\sup_M u \leq 1$, yielding the desired inequality.

Observe that

$$\Delta_{\tilde{\omega}} \log u \geq nu^{\frac{1}{n}} - n \geq \log u.$$

Suppose there is a point $p \in M$ such that $u(p) = \sup_M u = 1$. By the strong maximum principle, we have $\log u \equiv 0$ in M and $\tilde{\omega}^n = \omega^n$. Then (3.2) yields $\operatorname{tr}_{\tilde{\omega}} \omega = n$. Consequently, we have $\tilde{\omega} = \omega$ thanks to the equality case of the arithmetic-geometric mean inequality. $\square$

Remark 3.2. Under the assumptions of Lemma 3.1, we have

$$F = \log \frac{\tilde{\omega}^n}{\omega^n} \geq 0.$$

Moreover, if $F(p) = 0$ at a point $p \in M$, then $\tilde{\omega}^n = \omega^n$ at p , hence $\tilde{\omega} = \omega$ on M .

Lemma 3.3. Under the assumptions of Lemma 3.1, suppose in addition that $\tilde{\omega} = \omega + \sqrt{-1} \partial \bar{\partial} \varphi$ for some $\varphi \in C^\infty(M)$ satisfying $\|\varphi\|_{C^2(M, \omega)} \leq A$. Then there exists a constant $C = C(n, A) > 1$ such that

$$C^{-1} \omega \leq \tilde{\omega} \leq C \omega.$$

Proof. Fix a point $x \in M$, and choose ω -normal coordinates in which $\tilde{\omega}$ is diagonal. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $\tilde{\omega}$ relative to ω . The C^2 -bound on φ gives

$$\lambda_j \leq \Lambda \quad \text{for all } j,$$

where $\Lambda > 1$ depends only on A . On the other hand, Lemma 3.1 gives

$$\prod_{j=1}^n \lambda_j = \frac{\tilde{\omega}^n}{\omega^n} \geq 1.$$

Hence, for every j ,

$$\lambda_j = \frac{\prod_{k=1}^n \lambda_k}{\prod_{k \neq j} \lambda_k} \geq \Lambda^{-(n-1)}.$$

This proves the metric comparison. $\square$

4. CscK metrics on a complete Kähler–Einstein manifold

4.1. Asymptotic rigidity and motivating examples

Let (M, ω) be a complete Kähler–Einstein manifold with scalar curvature $-n$, and let

$$\omega_\varphi := \omega + \sqrt{-1} \partial \bar{\partial} \varphi$$

be a complete cscK perturbation of ω , where $\varphi \in C^\infty(M)$. Suppose

$$\begin{cases} \omega_\varphi^n = e^F \omega^n, \\ \Delta_{\omega_\varphi} F = n - \text{tr}_{\omega_\varphi} \omega, \end{cases} \quad (4.1)$$

for some $F \in C^\infty(M)$. Without loss of generality, by adding a constant to φ , we normalize $\sup_M \varphi = 0$. We will keep this convention throughout this section.

On compact Kähler manifolds, cscK metrics cohomologous to a Kähler–Einstein metric are automatically Kähler–Einstein:

Proposition 4.1 (*A well-known fact*). *Let (M, ω) be a compact Kähler manifold. Suppose that $2\pi c_1(M) = \lambda[\omega]$ for some constant λ , where $c_1(M)$ is the first Chern class of M . If ω is a cscK metric, then ω is Kähler–Einstein.*

Question 1.1 asks for a non-compact analogue of this rigidity statement. In the complete non-compact setting, such a conclusion can no longer follow from compact cohomological arguments alone, so one must impose additional information. Accordingly, a natural strategy is to seek sufficient conditions formulated at infinity. Relative to the reference Kähler–Einstein metric ω , the pair (F, φ) satisfies the coupled elliptic system (4.1). This makes asymptotic information particularly relevant: analytic behavior at infinity is precisely the type of information that can be translated into the interior through the elliptic maximum principle. From this perspective, it is natural to begin with asymptotic conditions as candidates for rigidity.

For this purpose, we distinguish two asymptotic behaviors for a cscK perturbation ω_φ :

(1) (*Weak asymptotic condition*) the volume ratio satisfies

$$\frac{\omega_\varphi^n}{\omega^n} \rightarrow 1 \quad \text{at infinity};$$

(2) (*Strong asymptotic condition*) for any $p \in M$, the metric perturbation satisfies

$$\lim_{r \rightarrow \infty} \|\varphi\|_{C^2(M \setminus B_\omega(p, r), \omega)} = 0.$$

The strong condition clearly implies the weak one.

Since a complete Einstein manifold with positive Einstein constant is compact by Myers' theorem, the non-compact Kähler–Einstein setting reduces to two cases: the Calabi–Yau case and the negatively curved case. We begin with the Calabi–Yau case, where rigidity is particularly transparent: weak asymptotic control of the volume ratio already suffices.

Proposition 4.2. *Let (M, ω) be a complete Calabi–Yau manifold. Suppose $\tilde{\omega}$ is a cscK metric on M with $R(\tilde{\omega}) = 0$, and that $\tilde{\omega}^n = e^F \omega^n$ for some $F \in C^\infty(M)$. If, for any point $p \in M$,*

$$\lim_{r \rightarrow \infty} \|F\|_{C^0(M \setminus B(p, r))} = 0,$$

then $\tilde{\omega}$ is Ricci-flat.

Proof. Since $\text{Ric}(\tilde{\omega}) = \text{Ric}(\omega) - \sqrt{-1} \partial \bar{\partial} F$ and $\text{Ric}(\omega) = 0$, tracing with respect to $\tilde{\omega}$ gives

$$0 = R(\tilde{\omega}) = \text{tr}_{\tilde{\omega}} \text{Ric}(\tilde{\omega}) = -\Delta_{\tilde{\omega}} F,$$

which implies F is harmonic. As $F \rightarrow 0$ at infinity, the maximum principle yields $F \equiv 0$, hence $\tilde{\omega}^n = \omega^n$ and $\text{Ric}(\tilde{\omega}) = \text{Ric}(\omega) = 0$. $\square$

We now turn to the negatively curved case. The cscK equation with $R(\omega_\varphi) = R(\omega)$ already implies that $F - \varphi$ is ω_φ -harmonic. As a first rigidity result, we impose a stronger asymptotic condition which forces this harmonic function to vanish at infinity.

Proposition 4.3. *Let (M, ω) be a complete Kähler–Einstein manifold with negative scalar curvature. Suppose there exists $\varphi \in C^\infty(M)$ such that $\omega_\varphi = \omega + \sqrt{-1} \partial \bar{\partial} \varphi$ defines a complete cscK metric with $R(\omega_\varphi) = R(\omega) = -n$. If, for every $p \in M$,*

$$\lim_{r \rightarrow \infty} \|\varphi\|_{C^2(M \setminus B(p, r))} = 0,$$

then $\varphi \equiv 0$ and $\omega_\varphi = \omega$.

Proof. It follows from (4.1) that

$$\Delta_{\omega_\varphi}(F - \varphi) = 0.$$

Since $\lim_{r \rightarrow \infty} \|\varphi\|_{C^2(M \setminus B(p, r))} = 0$ for any $p \in M$, $\varphi \rightarrow 0$ at infinity. Moreover, the first equation in (4.1) gives

$$F = \log \det (I + \omega^{-1} \sqrt{-1} \partial \bar{\partial} \varphi) \rightarrow 0.$$

Hence $F - \varphi \rightarrow 0$ at infinity. The maximum principle then implies that $F - \varphi \equiv 0$ on M . Substituting this into (4.1) yields

$$(\omega + \sqrt{-1} \partial \bar{\partial} \varphi)^n = e^\varphi \omega^n. \quad (4.2)$$

Since $\varphi \rightarrow 0$ at infinity, the function φ is bounded on M . If $\sup_M \varphi > 0$, then φ attains a positive maximum at an interior point. At that point, we have

$$1 \geq \frac{(\omega + \sqrt{-1} \partial \bar{\partial} \varphi)^n}{\omega^n} = e^{\sup_M \varphi},$$

which is a contradiction. This implies $\sup_M \varphi \leq 0$. Similarly, an interior negative minimum would give

$$e^{\inf_M \varphi} \geq 1,$$

again a contradiction. Thus $\varphi \equiv 0$, and consequently $\omega_\varphi = \omega$. $\square$

4.2. Parabolicity and rigidity of complete Kähler–Einstein metrics

In particular, Proposition 4.3 shows that if a Kähler–Einstein metric and a cscK metric are asymptotically equivalent in the strong sense, then they must coincide. However, the condition

$$\|\varphi\|_{C^2} \rightarrow 0$$

already yields C^0 -decay of φ and F , and hence provides explicit C^0 -control at infinity. From the analytic point of view, it is therefore natural to ask whether this asymptotic C^0 -information of F and φ can be weakened to a more flexible integral condition, preferably of L^2 -type. Once we move from pointwise decay to global integral control, the distinction between parabolic and non-parabolic manifolds becomes essential.

Definition 4.4 (*Parabolicity*). A complete Riemannian manifold (M, g) is called *non-parabolic* if it admits a positive minimal Green’s function. Otherwise, it is called *parabolic*.

A basic consequence of parabolicity is that every subharmonic function bounded from above must be constant (see, for instance, [21]). In our setting, this applies directly to the perturbation potential φ when the background Kähler–Einstein manifold is parabolic. It follows from Lemma 3.1 that

$$\Delta_\omega \varphi = \operatorname{tr}_\omega \omega_\varphi - n \geq n (\omega_\varphi^n / \omega^n)^{1/n} - n \geq 0. \quad (4.3)$$

Hence we obtain the following rigidity statement in the parabolic case.

Proposition 4.5. *Under the same assumptions as in Theorem 1.2, if (M, ω) is parabolic, then $\varphi \equiv 0$. In particular, ω_φ is Kähler–Einstein.*

Remark 4.6 (*Quadratic volume growth implies parabolicity*). A standard volume-growth criterion asserts that if

$$\int_0^\infty \frac{r}{\operatorname{Vol}_g(B(p, r))} dr = \infty,$$

then (M, g) is parabolic. In particular, if $\operatorname{Vol}_g(B(p, r)) \leq Cr^2$ for all sufficiently large r , then the integral diverges and (M, g) is parabolic; see also [13].

We now turn to the non-parabolic case. For the rigidity argument below, we impose the stronger quantitative assumption that

$$\lambda_1(-\Delta_\omega) := \inf_{0 \neq f \in W_0^{1,2}(M)} \frac{\int_M |\nabla f|_\omega^2 \omega^n}{\int_M |f|^2 \omega^n} > 0.$$

This implies (M, ω) is non-parabolic. Moreover, under the additional assumption $\|\varphi\|_{C^2(M, \omega)} < \infty$, the perturbation metric ω_φ is uniformly equivalent to the reference metric ω thanks to Lemma 3.3. We first show the following Liouville-type statement.

Lemma 4.7. *Let (M, g) be a complete Kähler manifold satisfying $\lambda_1(-\Delta_g) > 0$. Let $u \in C^\infty(M)$ be g -harmonic. Suppose that, for some point $p \in M$, there exist constants $C_0 > 0$ and $0 < \delta < 2\sqrt{\lambda_1(-\Delta_g)}$ such that for every $R \geq 1$, we have*

$$\int_{B_g(p, 2R) \setminus B_g(p, R)} u^2 dV_g \leq C_0 e^{2\delta R}. \quad (4.4)$$

Then $u \equiv 0$.

Proof. Fix a number $\alpha \in (\frac{\delta}{2}, \sqrt{\lambda_1(-\Delta_g)})$, and let $\rho(x) := d_g(p, x)$ be the distance function. Choose a nonincreasing function $\chi \in C^\infty([0, \infty))$ such that

$$0 \leq \chi \leq 1, \quad \chi \equiv 1 \text{ on } [0, 1], \quad \chi \equiv 0 \text{ on } [2, \infty), \quad |\chi'| \leq C.$$

For $R \geq 1$, define

$$\chi_R(x) := \chi\left(\frac{\rho(x)}{R}\right), \quad \eta_R(x) := \chi_R(x) e^{-\alpha \rho(x)}.$$

Then $\eta_R \in W_c^{1,\infty}(M)$, $\text{supp } \eta_R \subset B_g(p, 2R)$, and

$$\eta_R = e^{-\alpha \rho} \quad \text{on } B_g(p, R).$$

In particular,

$$\eta_R u, \eta_R^2 u \in W_0^{1,2}(M).$$

Since u is g -harmonic, its weak formulation gives

$$\int_M \langle \nabla u, \nabla \psi \rangle_g dV_g = 0 \quad \forall \psi \in W_0^{1,2}(M).$$

Taking $\psi = \eta_R^2 u$, we obtain

$$0 = \int_M \eta_R^2 |\nabla u|_g^2 dV_g + 2 \int_M \eta_R u \langle \nabla u, \nabla \eta_R \rangle_g dV_g.$$

On the other hand,

$$|\nabla(\eta_R u)|_g^2 = \eta_R^2 |\nabla u|_g^2 + u^2 |\nabla \eta_R|_g^2 + 2\eta_R u \langle \nabla u, \nabla \eta_R \rangle_g.$$

Combining the last two identities yields

$$\int_M |\nabla(\eta_R u)|_g^2 dV_g = \int_M u^2 |\nabla \eta_R|_g^2 dV_g.$$

By the variational characterization of $\lambda_1(-\Delta_g)$,

$$\lambda_1(-\Delta_g) \int_M (\eta_R u)^2 dV_g \leq \int_M |\nabla(\eta_R u)|_g^2 dV_g,$$

hence

$$\lambda_1(-\Delta_g) \int_M \eta_R^2 u^2 dV_g \leq \int_M u^2 |\nabla \eta_R|_g^2 dV_g.$$

Since

$$\nabla \eta_R = e^{-\alpha\rho} \nabla \chi_R - \alpha \chi_R e^{-\alpha\rho} \nabla \rho$$

and $|\nabla \rho|_g \leq 1$ almost everywhere, for every $\varepsilon > 0$ we have

$$|\nabla \eta_R|_g^2 \leq (1 + \varepsilon) \alpha^2 \eta_R^2 + C_\varepsilon e^{-2\alpha\rho} |\nabla \chi_R|_g^2.$$

Since $|\nabla \chi_R|_g \leq C/R$ and $\nabla \chi_R$ is supported in $A_R := B_g(p, 2R) \setminus B_g(p, R)$, we get

$$|\nabla \eta_R|_g^2 \leq (1 + \varepsilon) \alpha^2 \eta_R^2 + \frac{C_\varepsilon}{R^2} e^{-2\alpha\rho} \mathbf{1}_{A_R}.$$

Choose $\varepsilon > 0$ so small that $\lambda_1(-\Delta_g) - (1 + \varepsilon) \alpha^2 > 0$. Therefore,

$$\left(\lambda_1(-\Delta_g) - (1 + \varepsilon) \alpha^2 \right) \int_M \eta_R^2 u^2 dV_g \leq \frac{C_\varepsilon}{R^2} \int_{A_R} e^{-2\alpha\rho} u^2 dV_g.$$

Let $E_k := B_g(p, k+1) \setminus B_g(p, k)$ for $k \geq 1$. Since

$$E_k \subset B_g(p, k+1) \setminus B_g\left(p, \frac{k+1}{2}\right) = A_{(k+1)/2},$$

we have

$$\int_{E_k} u^2 dV_g \leq C_0 e^{\delta(k+1)}.$$

It follows from $A_R \subset \bigcup_{k=\lfloor R \rfloor}^{\lfloor 2R \rfloor} E_k$ and $\rho \geq k$ on E_k that

$$\begin{aligned} \int_{A_R} e^{-2\alpha\rho} u^2 dV_g &\leq \sum_{k=\lfloor R \rfloor}^{\lfloor 2R \rfloor} e^{-2\alpha k} \int_{E_k} u^2 dV_g \\ &\leq C_0 e^\delta \sum_{k=\lfloor R \rfloor}^{\lfloor 2R \rfloor} e^{-(2\alpha-\delta)k} \\ &\leq C_{\alpha,\delta} e^{-(2\alpha-\delta)R}. \end{aligned}$$

Consequently,

$$\left(\lambda_1(-\Delta_g) - (1 + \varepsilon) \alpha^2 \right) \int_M \eta_R^2 u^2 dV_g \leq \frac{C}{R^2} e^{-(2\alpha-\delta)R} \xrightarrow{R \rightarrow \infty} 0.$$

Since χ is nonincreasing, for each fixed $x \in M$ we have

$$\eta_R(x)^2 u(x)^2 \nearrow e^{-2\alpha\rho(x)} u(x)^2 \quad \text{as } R \rightarrow \infty.$$

By the monotone convergence theorem,

$$\int_M e^{-2\alpha\rho} u^2 dV_g = \lim_{R \rightarrow \infty} \int_M \eta_R^2 u^2 dV_g = 0.$$

Hence $u \equiv 0$ on M . $\square$

Lemma 4.8. *Under the same assumption as in Theorem 1.2. Suppose in addition that $\|\varphi\|_{C^2(M,\omega)} < \infty$. Then there exist constants $C > 0$ and $L > 1$, independent of $p \in M$ and R , such that, for every $R > 2L$,*

$$\int_{B_{\omega_\varphi}(p, 2R) \setminus B_{\omega_\varphi}(p, R)} |F - \varphi|^2 \omega_\varphi^n \leq C \int_{B_{\omega_\varphi}(p, 2R+L) \setminus B_{\omega_\varphi}(p, R-L)} |\varphi|^2 \omega_\varphi^n. \quad (4.5)$$

Moreover, for every $p \in M$ and every $\delta \geq 0$,

$$\int_{B_{\omega_\varphi}(p, 2R) \setminus B_{\omega_\varphi}(p, R)} |\varphi|^2 \omega_\varphi^n = O(e^{2\delta R}) \iff \int_{B_{\omega_\varphi}(p, 2R) \setminus B_{\omega_\varphi}(p, R)} |F - \varphi|^2 \omega_\varphi^n = O(e^{2\delta R}) \quad (4.6)$$

as $R \rightarrow \infty$.

Proof. Write again $u := F - \varphi$, $g := \omega_\varphi$, $dV_g := \omega_\varphi^n$. We first prove a local estimate. At any point, choose ω -normal coordinates in which g has eigenvalues $\lambda_1, \dots, \lambda_n$ relative to ω . Since $\log t \leq t - 1$ for every $t > 0$,

$$F = \sum_{j=1}^n \log \lambda_j \leq \sum_{j=1}^n (\lambda_j - 1) = \Delta_\omega \varphi.$$

It follows from Lemma 3.1 and $\varphi \leq 0$ that

$$0 \leq u \leq \Delta_\omega \varphi - \varphi. \quad (4.7)$$

Fix $x \in M$. By Lemma 3.3, there exists a constant $A \geq 1$, independent of x , such that

$$A^{-1}\omega \leq g \leq A\omega.$$

Let $r_x = d_\omega(x, \cdot)$, and choose a non-increasing function $\chi \in C^\infty([0, \infty))$ such that

$$0 \leq \chi \leq 1, \quad \chi \equiv 1 \text{ on } [0, 1], \quad \chi \equiv 0 \text{ on } [2, \infty).$$

Define $\eta_x := \chi(A^{-1/2}r_x)$. Then $\eta_x \equiv 1$ on $B_g(x, 1)$ and $\text{supp } \eta_x \subset B_g(x, L)$, where we may take $L = 3A$. Since $\text{Ric}(\omega) = -\omega$, Laplacian comparison gives a uniform upper bound for $\Delta_\omega r_x$ on the region $A^{1/2} \leq r_x \leq 2A^{1/2}$. Because $\chi' \leq 0$, it follows that $\Delta_\omega \eta_x \geq -C_\eta$ in the distributional sense, where C_η is independent of x . Using (4.7) and $\varphi \leq 0$, we obtain

$$\begin{aligned}
\int_{B_g(x,1)} u \, dV_g &\leq C \int_M \eta_x u \, \omega^n \\
&\leq C \int_M \eta_x (\Delta_\omega \varphi - \varphi) \, \omega^n \\
&= C \langle \Delta_\omega \eta_x, \varphi \rangle_\omega - C \int_M \eta_x \varphi \, \omega^n \\
&\leq C \int_{B_g(x,L)} |\varphi| \, dV_g.
\end{aligned}$$

Here the third line is understood in the distributional sense, and the last inequality follows precisely from $\Delta_\omega \eta_x \geq -C_\eta$ and $\varphi \leq 0$.

If $u \equiv 0$, there is nothing to prove. Otherwise, since $u \geq 0$ and $\Delta_g u = 0$, the strong maximum principle implies that $u > 0$ on M . Recall Harnack's inequality [31, Corollary 3.3],

$$\sup_{B_g(x,1)} u \leq C \inf_{B_g(x,1)} u, \quad (4.8)$$

where C depends only on the dimension and the lower bound of $\text{Ric}(g)$. Therefore, by (4.8), we have

$$\begin{aligned}
\int_{B_g(x,1)} u^2 \, dV_g &\leq \frac{C}{\text{Vol}_g(B_g(x,1))} \left(\int_{B_g(x,1)} u \, dV_g \right)^2 \\
&\leq \frac{C}{\text{Vol}_g(B_g(x,1))} \left(\int_{B_g(x,L)} |\varphi| \, dV_g \right)^2 \\
&\leq C \frac{\text{Vol}_g(B_g(x,L))}{\text{Vol}_g(B_g(x,1))} \int_{B_g(x,L)} |\varphi|^2 \, dV_g \\
&\leq C \int_{B_g(x,L)} |\varphi|^2 \, dV_g,
\end{aligned}$$

where the last inequality follows from Bishop–Gromov volume comparison.

Let $A_R := B_g(p, 2R) \setminus B_g(p, R)$. Choose a maximal 1-separated set $\{x_j\}_{j=1}^{N_R} \subset A_R$. Then

$$A_R \subset \bigcup_j B_g(x_j, 1),$$

while the balls $B_g(x_j, \frac{1}{2})$ are pairwise disjoint. For $y \in B_g(x_j, L)$,

$$B_g(x_j, \tfrac{1}{2}) \subset B_g(y, L + \tfrac{1}{2}) \subset B_g(x_j, 2L + \tfrac{1}{2}).$$

Hence Bishop–Gromov volume comparison gives

$$\text{Vol}_g(B_g(x_j, \tfrac{1}{2})) \geq C \text{Vol}_g(B_g(y, L + \tfrac{1}{2})).$$

The family $\{B_g(x_j, L)\}_j$ has uniformly bounded multiplicity. Indeed, let $I(y) := \{j; y \in B_g(x_j, L)\}$. Since the balls $B_g(x_j, \frac{1}{2})$ are disjoint, it follows that

$$\text{Vol}_g(B_g(y, L + \tfrac{1}{2})) \geq \sum_{j \in I(y)} \text{Vol}_g(B_g(x_j, \tfrac{1}{2})) \geq \#I(y) \cdot C \text{Vol}_g(B_g(y, L + \tfrac{1}{2}))$$

Therefore, for some N depending only on the dimension, L and the lower bound of $\text{Ric}(g)$, we have $\#I(y) \leq N$. Consequently,

$$\begin{aligned} \int_{A_R} u^2 dV_g &\leq \sum_j \int_{B_g(x_j, 1)} u^2 dV_g \\ &\leq C \sum_j \int_{B_g(x_j, L)} \varphi^2 dV_g \\ &\leq CN \int_{B_g(p, 2R+L) \setminus B_g(p, R-L)} \varphi^2 dV_g, \end{aligned}$$

which proves (4.5).

We now prove (4.6). Suppose first that

$$\int_{A_R} |\varphi|^2 dV_g = O(e^{2\delta R}).$$

For $R \geq 3L$, we have

$$(B_g(p, 2R+L) \setminus B_g(p, R-L)) \subset (A_{R-L} \cup A_{R+L}).$$

It follows from (4.5) that

$$\begin{aligned} \int_{A_R} |u|^2 dV_g &\leq C \left(\int_{A_{R-L}} |\varphi|^2 dV_g + \int_{A_{R+L}} |\varphi|^2 dV_g \right) \\ &\leq C' (e^{2\delta(R-L)} + e^{2\delta(R+L)}) \\ &\leq C'' e^{2\delta R}. \end{aligned}$$

Conversely, Lemma 3.1 and $\sup_M \varphi = 0$ give

$$u = F - \varphi = F + |\varphi| \geq |\varphi|,$$

and hence

$$\int_{A_R} |\varphi|^2 dV_g \leq \int_{A_R} |u|^2 dV_g.$$

This proves (4.6). $\square$

Corollary 4.9. Assume the hypotheses of Theorem 1.2 (ii). Then $\varphi \equiv 0$.

Proof. We use the notation introduced in the proof of Lemma 4.8. For R sufficiently large, by Lemma 4.8, we have

$$\int_{A_R} u^2 dV_g \leq C_1 e^{2\delta R}.$$

After increasing C_1 , the same inequality holds for all $1 \leq R$. Since $0 < \delta < 2\sqrt{\lambda_1(-\Delta_g)}$, Lemma 4.7 gives $u \equiv 0$. Thus $F = \varphi$. Lemma 3.1 gives $F \geq 0$, whereas $\varphi \leq 0$. Hence $F \equiv \varphi \equiv 0$. $\square$

5. The cscK metric on bounded strictly pseudoconvex domains

In this section, we study cscK metrics on bounded strictly pseudoconvex domains in $\mathbb{C}^n$. We will also briefly recall some general facts about the Bergman metric on bounded pseudoconvex domains. Note that bounded pseudoconvex domains admit complete Kähler–Einstein metrics of negative scalar curvature, see [25].

5.1. The Cheng–Yau metric

Let $\Omega \subset \mathbb{C}^n$ be a bounded strictly pseudoconvex domain with C^k -boundary, where $k \geq 2$. In [14], Cheng and Yau studied complete Kähler metrics arising from defining functions. In the present subsection, we consider metrics of the form

$$\omega_\rho := -\sqrt{-1} \partial \bar{\partial} \log(-\rho),$$

where ρ is a negative globally strictly plurisubharmonic defining function for $\Omega = \{\rho < 0\}$. Here, by globally strictly plurisubharmonic, we mean that ρ extends as a strictly plurisubharmonic function to a neighborhood of $\bar{\Omega}$. In particular,

$$\omega_\rho = -\frac{\sqrt{-1} \partial \bar{\partial} \rho}{\rho} + \frac{\sqrt{-1} \partial \rho \wedge \bar{\partial} \rho}{\rho^2} > 0,$$

so ω_ρ is a Kähler metric on Ω . Moreover, the second term shows that the boundary is at infinite ω_ρ -distance, and hence ω_ρ is complete.

For a C^2 defining function r , the Fefferman operator is defined by

$$J(r) := -\det \begin{pmatrix} r & r_{\bar{j}} \\ r_i & r_{i\bar{j}} \end{pmatrix}.$$

A direct computation gives

$$\det((\omega_r)_{i\bar{j}}) = \frac{J(r)}{(-r)^{n+1}}. \quad (5.1)$$

In particular, whenever r is smooth in Ω and ω_r is Kähler, one has $J(r) > 0$ in Ω , and (5.1) yields

$$\text{Ric}(\omega_r) = -(n+1)\omega_r - \sqrt{-1} \partial \bar{\partial} \log J(r). \quad (5.2)$$

Cheng and Yau [14] proved that the Dirichlet problem

$$\begin{cases} J(\rho_0) = 1 & \text{in } \Omega, \\ \rho_0 = 0 & \text{on } \partial\Omega, \end{cases} \quad (5.3)$$

admits a unique solution ρ_0 for which

$$\omega_0 := -\sqrt{-1} \partial \bar{\partial} \log(-\rho_0)$$

is a complete Kähler metric. By (5.2), this metric satisfies

$$\text{Ric}(\omega_0) = -(n+1)\omega_0, \quad R(\omega_0) = -n(n+1).$$

We refer to ρ_0 as the exact Cheng–Yau defining function and to ω_0 as the unique Kähler–Einstein metric.

For any other C^k -defining function ρ , there exists a positive function u such that $\rho_0 = u\rho$. Hence

$$J(\rho) = u^{-(n+1)} J(\rho_0) = u^{-(n+1)} \quad \text{on } \partial\Omega.$$

Multiplication of ρ by a positive constant does not change the associated metric ω_ρ , but it replaces u by a constant multiple of u . Thus such a constant rescaling yields $u|_{\partial\Omega} = 1$ only when $u|_{\partial\Omega}$ is constant. In what follows, we restrict attention to defining functions satisfying the boundary normalization $J(\rho) = 1$ on $\partial\Omega$.

Proposition 5.1. *Let $\Omega \subset \mathbb{C}^n$ be a bounded strictly pseudoconvex domain with C^2 boundary, and let $\rho \in C^\infty(\Omega) \cap C^4(\bar{\Omega})$ be a negative strictly plurisubharmonic defining function satisfying $J(\rho) = 1$ on $\partial\Omega$. If $\omega := -\sqrt{-1} \partial \bar{\partial} \log(-\rho)$ has constant scalar curvature, then ω is Kähler–Einstein.*

Proof. By the fundamental computations of curvature tensor by Cheng–Yau [14, (1.22)], we have

$$R_{i\bar{j}k\bar{l}}(\omega) = -(g_{i\bar{j}}g_{k\bar{l}} + g_{i\bar{l}}g_{k\bar{j}}) + O(|\rho|^{-1}) \quad \text{as } z \rightarrow \partial\Omega.$$

Together with $g^{i\bar{j}} = O(|\rho|)$ ([14, (1.2)]), this gives

$$R(\omega) = -n(n+1) + O(|\rho|) \rightarrow -n(n+1) \quad \text{as } z \rightarrow \partial\Omega. \quad (5.4)$$

Taking the trace of (5.2) with respect to ω , we obtain

$$R(\omega) = -n(n+1) - \Delta_\omega \log J(\rho). \quad (5.5)$$

Since $R(\omega)$ is constant, (5.4) implies $R(\omega) = -n(n+1)$. It follows from (5.5) that

$$\Delta_\omega \log J(\rho) = 0 \quad \text{in } \Omega.$$

Moreover, $J(\rho) \in C^2(\bar{\Omega})$ is positive in Ω and satisfies $J(\rho) = 1$ on $\partial\Omega$. Hence $\log J(\rho)$ extends continuously to $\bar{\Omega}$ and vanishes on $\partial\Omega$. The maximum principle therefore gives

$$\log J(\rho) \equiv 0.$$

Consequently, (5.2) implies that ω is Kähler–Einstein. In particular, since ω is complete, the uniqueness argument in [14, Theorem 4.4] shows that $\omega = \omega_0$ where ω_0 is the complete Kähler–Einstein metric constructed by Cheng and Yau. $\square$

5.2. The Bergman metric

Let $\Omega \subset \mathbb{C}^n$ be a bounded domain, and let $K_\Omega(z, w)$ denote its Bergman kernel. We write $K(z) := K_\Omega(z, z)$. The Bergman metric is defined by

$$\omega_B := \sqrt{-1} \partial \bar{\partial} \log K.$$

On a bounded strictly pseudoconvex domain with C^1 boundary, ω_B is a complete real-analytic Kähler metric [27].

Let $G(z)$ be the determinant of the Bergman metric. The Bergman invariant function is defined by $B(z) := \frac{G(z)}{K(z)}$. The transformation laws of G and K show that B is invariant under biholomorphic maps. Since $\log G = \log B + \log K$, one has

$$\text{Ric}(\omega_B) + \sqrt{-1} \partial \bar{\partial} \log B = -\omega_B. \quad (5.6)$$

The boundary behavior of the Bergman metric has been widely studied. A key observation is that, on a bounded pseudoconvex domain Ω with C^2 boundary, the Bergman metric ω_B behaves asymptotically like a Kähler–Einstein metric near a strictly pseudoconvex boundary point [15]. The problem of promoting such boundary asymptotics to a global Kähler–Einstein equation belongs to a broader conjectural picture proposed by Yau [38]. He asked that, for a bounded domain which is not a product domain, the Bergman metric is complete and Kähler–Einstein if and only if the domain is homogeneous. For strictly pseudoconvex domains, the Wong–Rosay theorem shows that a bounded homogeneous domain with C^2 boundary must be biholomorphic to the ball [29,35]. A particularly tractable case is therefore Cheng’s conjecture: *a smoothly bounded strictly pseudoconvex domain whose Bergman metric is Kähler–Einstein must be biholomorphic to the ball*. Fu–Wong and Nemirovski–Shafikov proved this conjecture in complex dimension two [19,26], and Huang–Xiao solved it in every complex dimension $n \geq 2$ [22, Theorem 1].

This leads naturally to a scalar-curvature version of Cheng’s conjecture: *does constant scalar curvature of the Bergman metric already characterize the ball?* The first step is to determine whether the constant-scalar-curvature condition forces the Bergman metric to be Kähler–Einstein. The following proposition gives an affirmative answer and, together with Huang–Xiao’s theorem, yields the ball characterization for bounded strictly pseudoconvex domains with smooth boundary.

Proposition 5.2. *Let $\Omega \subset \mathbb{C}^n$ be a bounded strictly pseudoconvex domain with C^2 boundary. If its Bergman metric ω_B has constant scalar curvature, then ω_B is Kähler–Einstein. Moreover, if $\partial\Omega \in C^\infty$, then Ω is biholomorphic to the unit ball.*

Proof. By [23, Corollary 2], at every boundary point $p \in \partial\Omega$ one has

$$\lim_{z \rightarrow p} R(\omega_B)(z) = -n, \quad \lim_{z \rightarrow p} B(z) = \frac{(n+1)^n \pi^n}{n!}. \quad (5.7)$$

Since $R(\omega_B)$ is constant, the first limit gives $R(\omega_B) \equiv -n$. Taking the trace of (5.6) with respect to ω_B , we obtain

$$R(\omega_B) + \Delta_{\omega_B} \log B = -n.$$

It follows that

$$\Delta_{\omega_B} \log B = 0 \quad \text{in } \Omega.$$

The second limit in (5.7) shows that $\log B$ extends continuously to $\bar{\Omega}$ with a constant boundary value. By the weak maximum principle, we conclude that $\log B$ is constant on $\bar{\Omega}$. Equation (5.6) therefore implies that ω_B is Kähler–Einstein. If $\partial\Omega \in C^\infty$, the final assertion follows from [22, Theorem 1]. $\square$

Data availability

No data was used for the research described in the article.

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